

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q22-2035

Spatsum Lornex 138 kV
March 21, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2035. The Interconnection Customer proposes a 290.0 MW solar photovoltaic facility interconnecting at the Spatsum Lornex 138 kV substation in the Kentucky area. The request was received on March 17, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 467 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 13 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$38,689,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Sample Rd to Tomahawk-Atc 138kV L
Contingency: Spatsum Lornex to Latah Jct 138kV
Loading: 139.2% (Rating: 225 MVA)
DFAX: 38.6%
Pre-project loading: 89.6%
2. Facility: Welfare to Kettle 138kV line
Contingency: Spatsum Lornex to Latah Jct 138kV
Loading: 139.0% (Rating: 256 MVA)
DFAX: 38.6%
Pre-project loading: 95.3%
3. Facility: Alibates to Latah Jct 138kV
Contingency: Harvest Power Orlando to Frontier 138kV L
Loading: 134.0% (Rating: 495 MVA)
DFAX: 61.4%
Pre-project loading: 98.1%
4. Facility: Spatsum Lonex to Latah Junction 138kV line
Contingency: Harvest Power Orlando to Frotier Substation 138kV
Loading: 133.0% (Rating: 508 MVA)
DFAX: 61.4%
Pre-project loading: 97.9%
5. Facility: Kettle to Country Club 2 138kV
Contingency: Latah Jct to Spatsum Lornex 138kV
Loading: 131.1% (Rating: 347 MVA)
DFAX: 53.5%
Pre-project loading: 86.3%

6. Facility: Spatsum Lornex to Country Club 2 Substation 138kV
Contingency: Spatsum Lornex to Latah Jct 138kV Line
Loading: 130.2% (Rating: 355 MVA)
DFAX: 53.5%
Pre-project loading: 86.4%
7. Facility: Tomahawk-Atc Substation to Welfare Substation 138kV
Contingency: Spatsum Lornex Substation to Latah Junction 138kV
Loading: 127.4% (Rating: 265 MVA)
DFAX: 38.6%
Pre-project loading: 85.2%
8. Facility: Sample Road to Monrovia 138kV line
Contingency: Latah Jct to Spatsum Lornex 138kV
Loading: 127.1% (Rating: 230 MVA)
DFAX: 31.2%
Pre-project loading: 87.7%
9. Facility: East Broad to Harvest Power Orlando 138kV
Contingency: Spatsum Lornex to Latah Jct 138kV L
Loading: 119.0% (Rating: 278 MVA)
DFAX: 30.8%
Pre-project loading: 86.9%
10. Facility: Harvest Power Orlando to Frontier 138kV
Contingency: Spatsum Lornex to Latah Jct 138kV
Loading: 115.4% (Rating: 488 MVA)
DFAX: 47.9%
Pre-project loading: 86.9%
11. Facility: Juliette to Alibates 138kV
Contingency: Action to Alibates 138kV Line
Loading: 109.8% (Rating: 368 MVA)
DFAX: 41.1%
Pre-project loading: 77.5%
12. Facility: N City to Boonville 138kV
Contingency: Latah Jct to Spatsum Lornex 138kV line
Loading: 108.5% (Rating: 195 MVA)
DFAX: 10.2%
Pre-project loading: 93.3%
13. Facility: Arverne to Tyrannena 138kV
Contingency: Keifer-Jackson to Beaver Dam East 138kV
Loading: 102.3% (Rating: 201 MVA)
DFAX: 7.1%
Pre-project loading: 92.1%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Sample Rd to Tomahawk-Atc 138kV L	Rebuild line with higher-capacity conductor	\$13,894,000	40% (\$5,543,000)
Welfare to Kettle 138kV L	Rebuild line with higher-capacity conductor	\$12,530,000	40% (\$4,999,000)
Alibates to Latah Jct 138kV	Rebuild line with higher-capacity conductor	\$10,287,000	34% (\$3,474,000)
Spatsum Lonex to Latah Junction 138kV line	Rebuild line with higher-capacity conductor	\$6,314,000	34% (\$2,132,000)
Kettle to Country Club 2 138kV	Rebuild line with higher-capacity conductor	\$9,234,000	43% (\$3,980,000)
Spatsum Lornex to Country Club 2 Substation 138kV	Rebuild line with higher-capacity conductor	\$6,880,000	43% (\$2,965,000)
Tomahawk-Atc Substation to Welfare Substation 138kV	Rebuild line with higher-capacity conductor	\$6,264,000	40% (\$2,499,000)
Monrovia to Sample Rd 138kV	Rebuild line with higher-capacity conductor	\$7,238,000	32% (\$2,282,000)
East Broad to Harvest Power Orlando 138kV	Rebuild line with higher-capacity conductor	\$7,848,000	31% (\$2,454,000)
Harvest Power Orlando to Frontier 138kV	Rebuild line with higher-capacity conductor	\$17,878,000	31% (\$5,619,000)
Alibates to Juliette 138kV	Reconductor line	\$4,064,000	37% (\$1,489,000)
North City to Boonville 138kV line	Reconductor line	\$6,361,000	18% (\$1,155,000)
Arverne to Tyrana 138kV	Reconductor line	\$1,002,000	10% (\$98,000)
	Total allocated to project		\$38,689,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.